Vertiport Planning in Chicago

Mathematical model, working document v1.1

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Abstract

This document states the optimization model behind the project. The framework (facility location with demand that responds to travel times, and a distributionally robust demand function) comes from Wang, Jacquillat and Vaze (2022), "Vertiport Planning for Urban Aerial Mobility: An Adaptive Discretization Approach," *M&SOM* 24(6):3215–3235. The project uses a simplified version without the operations layer and asks two questions.
RQ1: What changes when customer expectations are estimated per origin–destination pair from observed rideshare fares and times, rather than set uniformly?
RQ2: How robust is the network to time-of-day demand, and what does a network planned for a single period lose over a year?

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1 Overview

The model has two layers. The *precomputation* layer (Python) turns trip records into travel times and switching shares for every possible routing. All nonlinear relations are computed there. The *optimization* layer, a mixed-integer linear program solved with Gurobi, then chooses vertiport sites and routes each origin–destination flow. All reported runs maximize air-taxi passengers; the profit version uses the margins of Section 4.5 as weights. Figure 1 shows the pipeline.

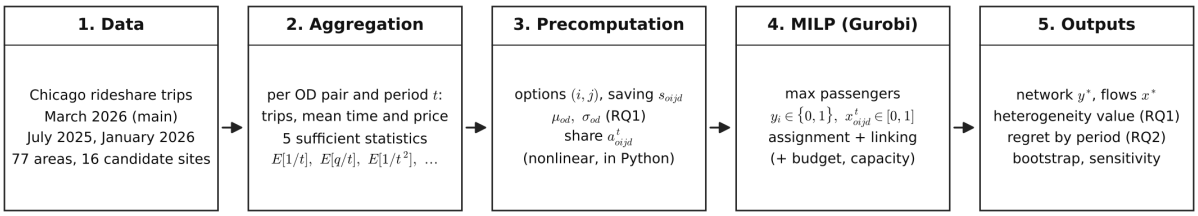


Figure 1: From trip records to the network and the RQ1/RQ2 metrics. t: ride time; q: ride price; s: relative time saving; a: switching share.

Figure 2 shows the choice for one origin–destination (OD) pair in one period: each traveler stays on the ground (rideshare) or flies through an open vertiport pair (i, j).

2 Sets and indices

Symbol	Index	Meaning
$\mathcal{Z}$	o, d	The 77 Chicago community areas (origins and destinations).
$\mathcal{V} \subseteq \mathcal{Z}$	i, j	Candidate sites. Main runs: 16 (O'Hare, Midway and Vertiport Chicago at exact locations, five high-demand areas, eight areas with cluster centers of Volakakis and Mahmassani 2024). Robustness runs: all 77 areas.
$\mathcal{T}$	t	Periods $\{\text{am}, \text{mid}, \text{pm}, \text{oth}, \text{we}\}$: weekday 7–10, 10–16, 16–19, other hours; weekend.
$\mathcal{P} \subseteq \mathcal{Z} \times \mathcal{Z}$	(o, d)	OD pairs kept per period: $o \neq d$, mean ground time ≥ 20 min, ≥ 1 trip per day.
$\mathcal{K}_{od} \subseteq \mathcal{V} \times \mathcal{V}$	(i, j)	Useful vertiport pairs for (o, d) (Section 4.2).

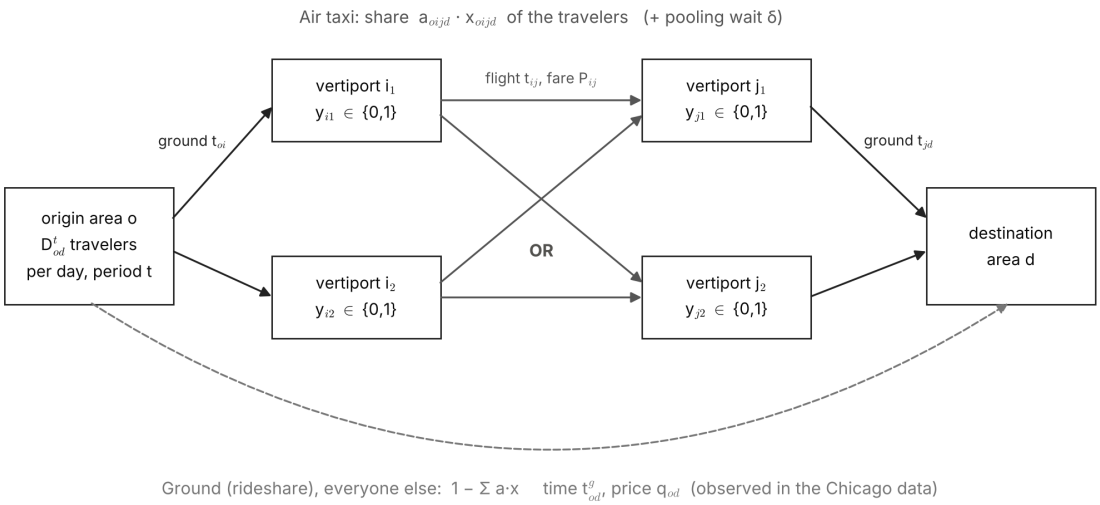


Figure 2: Routing options for OD pair (o, d) in period t. Black: ground legs to and from vertiports; grey: flight legs; dashed: travelers who stay on the ground.

3 Data and parameters

3.1 Observed in the Chicago rideshare data

Source: City of Chicago, Transportation Network Providers trips (Socrata `6dvr-xwnh`). March 2026 gives the main results; July 2025 and January 2026 the seasonal check. Trips are aggregated by pickup area, drop-off area and period, one day at a time. Removed: shared trips, trips under 60 seconds or over 3 hours, and trips missing a fare or an area. Ground time is `trip_seconds` (timestamps rounded to 15 minutes). Ground price is `fare` + `additional_charges` (fares rounded to \$2.50).

Symbol	Meaning	Unit
D^t_{od}	Rideshare trips $o \to d$ per day of period type t	trips/day
$t^{g,t}_{od}$	Mean ground time $o \to d$ in period t	min
q_k, t_k	Price and ground time of trip k	\$, min
$m^{(1..5),t}_{od}$	Sufficient statistics $\mathbb{E}[1/t]$, $\mathbb{E}[q/t]$, $\mathbb{E}[1/t^2]$, $\mathbb{E}[q/t^2]$, $\mathbb{E}[q^2/t^2]$ (Section 5)	various
ℓ_{od}, ℓ_{ij}	Straight-line (haversine) distance between centroids or sites	miles

3.2 Assumed (with sensitivity ranges)

Values come from public sources; the paper's online appendix was not available.

Symbol	Meaning	Default	Range
v	eVTOL cruise speed (240 km/h for NYC)	200 km/h	150–240
t_{oh}	Fixed time per flight (boarding, takeoff, landing)	10 min	6–20
$\bar{\delta}$	Pooling wait (endogenous in the paper)	5 min	2–10
t_0	Ground leg within the vertiport's own zone	5 min	3–8
h	Transfer ("hassle") penalty added to air-taxi time	0 min	0–15
$s_{\min}$	Minimum relative time saving for an option	0.15	0–0.15
$t_{\max}$	Maximum ground leg to or from a vertiport	25 min	15–30
k	Nearest vertiports considered at each trip end	3	2–5
μ, σ	Uniform expectations (NYC only; Chicago: Section 5)	0.40, 0.20	μ: 0.33–0.40
P_0, r, b_0	Base fare, rate per mile, median value of time	see Table 1	–
CV_b	Coefficient of variation of the value of time	0.84	–
κ, c^g, g_i	Flight cost per mile, ground-leg cost, daily site cost (profit version)	not used	–
w_t	Days per year of period type t	250 (weekday), 115 (weekend)	–

Scenario	P_0	r (/mile)	b_0 (/hour)	Basis
Launch	\$15	\$6.00	\$62.00	near-term fare estimates; airport and business travelers
Launch, pessimistic	\$15	\$12.00	\$62.00	current helicopter shuttle prices
Adopter (main)	\$15	\$3.50	\$31.00	operator targets; air-taxi survey of Boddupalli et al. (2024)
Mature	\$15	\$2.50	\$20.10	long-run fare estimates; USDOT personal local travel

Table 1: Fare and value-of-time scenarios (2024 dollars). The rate applies to the straight-line OD distance ℓ_{od}; the value of time is lognormal (median b_0, $\text{CV}_b = 0.84$).

4 Precomputation

4.1 Ground and flight times

Ground legs. For zones u and v in period t:

$$t^t_{uv} = \begin{cases} t_0 & \text{if } u = v, \\ \text{observed mean } t^{g,t}_{uv} & \text{if } \geq 2 \text{ trips per day}, \\ \alpha^t + \beta^t \, \ell_{uv} & \text{otherwise (linear fit on pairs with } \geq 5 \text{ trips per day}). \end{cases} \tag{1}$$

Flight legs. With ℓ_{ij} in km:

$$t^f_{ij} = \frac{60 \, \ell_{ij}}{v} + t_{\text{oh}}. \tag{2}$$

4.2 Useful vertiport pairs

Let $N^{\text{acc},t}(o)$ be the k sites closest to o by ground time, with $t^t_{oi} \leq t_{\max}$, and $N^{\text{egr},t}(d)$ the k closest to d, with $t^t_{jd} \leq t_{\max}$. Then

$$\mathcal{K}^t_{od} = \left\{ (i,j) \in N^{\text{acc},t}(o) \times N^{\text{egr},t}(d) \;:\; i \neq j \right\}. \tag{3}$$

4.3 Air-taxi trip time and time saving

As in equation (5) of the paper, with a fixed pooling wait instead of endogenous pooling and detour times:

$$\tau^t_{oijd} = t^t_{oi} + \bar{\delta} + t^f_{ij} + t^t_{jd} + h, \tag{4}$$

$$s^t_{oijd} = 1 - \frac{\tau^t_{oijd}}{t^{g,t}_{od}} \qquad \text{(relative time saving)}. \tag{5}$$

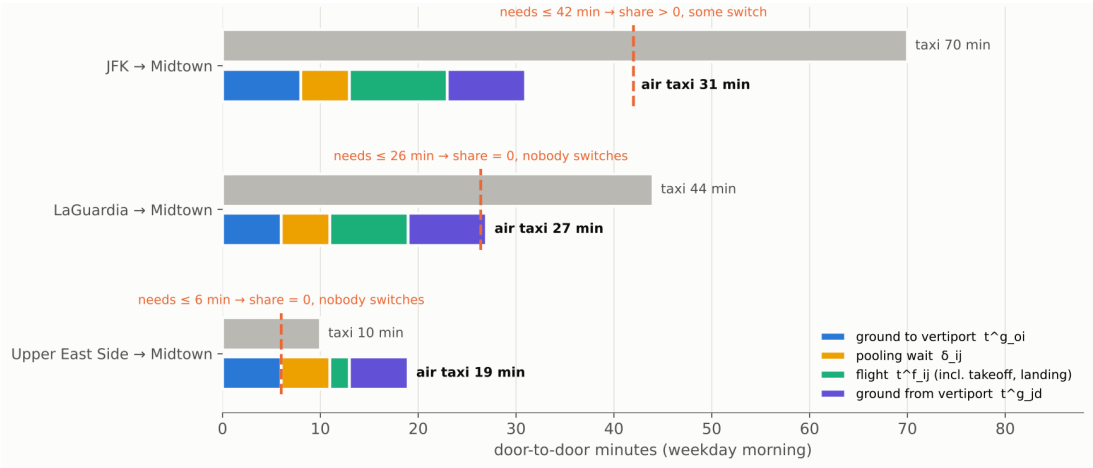


Figure 3: Air-taxi trip versus taxi, NYC validation example: observed taxi times, illustrative air-taxi components, dashed line at the $\mu = 0.40$ threshold. Only long trips leave a margin after the fixed components.

4.4 Demand: the share of travelers who switch

As in Section 4.4 of the paper, a traveler on (o, d) compares utilities $V - P_{od} - b\,\tau$ (air taxi) and $V - q_{od} - b\,t^g_{od}$ (ground). She flies if and only if

$$\tau \le (1-\theta)\,t^g_{od} \quad \Longleftrightarrow \quad \theta \le s, \qquad \text{where} \quad \theta = \frac{P_{od} - q_{od}}{b\,t^g_{od}} \tag{6}$$

is the *required relative time saving*: the fare premium, converted to minutes, divided by ground time. With times in minutes, b is in dollars per minute. θ varies across travelers and trips, with mean μ and standard deviation σ; the share who fly is $\Pr(\theta \le s)$. Two versions are used: the paper's distributionally robust (DRO) share, and an exact share under stated distributions, which gives the main results.

DRO share (Proposition 1 of the paper). If only μ and σ are known, the worst case over all such distributions follows from Cantelli's one-sided Chebyshev inequality:

$$\inf_{F:\ \mathbb{E}_F\theta=\mu,\ \mathrm{Var}_F\theta=\sigma^2} \Pr_F(\theta \le s) = \begin{cases} \dfrac{(s-\mu)^2}{\sigma^2 + (s-\mu)^2} & s \ge \mu, \\ 0 & s < \mu. \end{cases} \tag{7}$$

With $\tilde\tau = (s-\mu)/\sigma$, the paper's share is

$$a^t_{oijd} = \frac{\tilde\tau^2}{1+\tilde\tau^2} \ \text{ if } \tilde\tau > 0, \qquad a^t_{oijd} = 0 \text{ otherwise}, \qquad \tilde\tau = \frac{s^t_{oijd} - \mu_{od}}{\sigma_{od}}. \tag{8}$$

For NYC, $\mu_{od} = \mu$ and $\sigma_{od} = \sigma$ on all pairs. For Chicago, both are estimated from data (Section 5). The bound is zero whenever $s \le \mu$. Since $\mu_{od} > 1$ on most Chicago pairs in most scenarios, it predicts almost no demand, so the exact share gives the main results. Both shares are precomputed, so the optimization stays linear. In the paper, τ depends on pooling and detour decisions, so the S-curve sits inside the model and needs adaptive discretization.

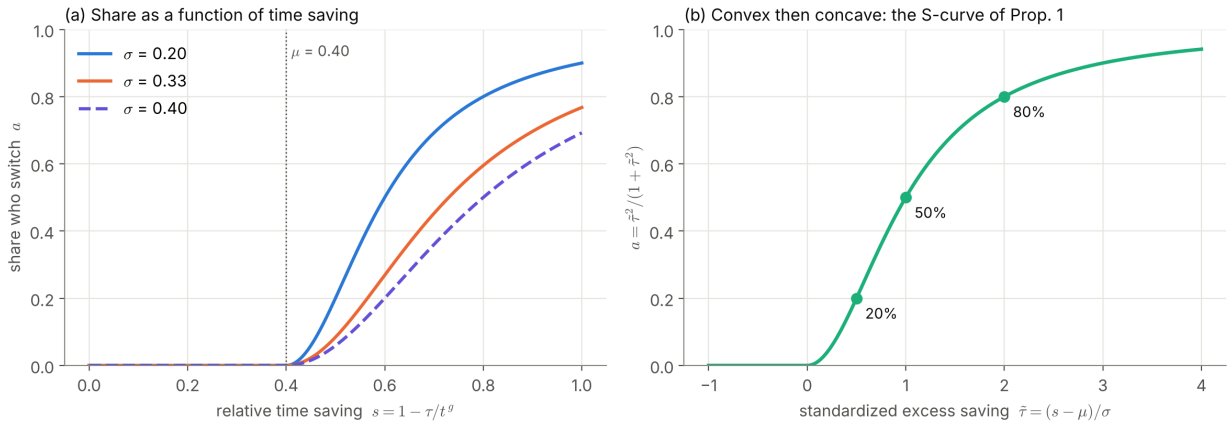


Figure 4: The DRO share (8). (a) Nobody switches until the saving exceeds μ; a larger σ flattens the curve. (b) In standardized units the curve is convex, then concave: 20%, 50% and 80% switch at $\tilde\tau = 0.5, 1, 2$.

Exact share (main results). Let $X = (P_{od} - q)/t$ be the per-minute fare premium, so $\theta = X/b$: a traveler takes an option with saving s if and only if $X \le s\,b$. Assume X is normal on each OD pair, with moments from Section 5, and b is lognormal (median b_0, $\mathrm{CV}_b = 0.84$), independent of X. Then

$$a^t_{oijd} = \mathbb{E}_b\left[\Phi\left(\frac{s^t_{oijd}\,b - \mathbb{E}[X]}{\mathrm{sd}(X)}\right)\right], \tag{9}$$

computed by 24-point Gauss–Hermite quadrature over $\ln b$. Options require $s^t_{oijd} \ge s_{\min} = 0.15$. This removes switching driven by price alone, when surged ground fares exceed the air-taxi fare with almost no time saved.

4.5 Fares and margins

The fare is door to door on the OD distance; the operator pays for the flight and both ground legs:

$$P_{od} = P_0 + r\,\ell_{od}, \tag{10}$$

$$c_{oijd} = \kappa\,\ell_{ij} + c^g\,(t^t_{oi} + t^t_{jd}) \qquad \text{(flight cost + ground-leg cost)}, \tag{11}$$

$$\pi_{oijd} = P_{od} - c_{oijd} \qquad \text{(margin per air-taxi passenger)}. \tag{12}$$

With $\pi_{oijd} = 1$ on all options, profit becomes passengers carried. Every reported run (NYC and Chicago) uses this objective, so κ, c^g and g_i are not calibrated.

5 Customer expectations from trip data (RQ1)

For trip k on (o, d) in period t, with price q_k and ground time t_k:

$$\theta_k = \frac{P_{od} - q_k}{b\,t_k}. \tag{13}$$

Sufficient statistics. Downloading every trip for each scenario would be slow. Instead, the City's data server returns five averages per (o, d, t):

$$m_1 = \mathbb{E}\left[\tfrac{1}{t}\right], \quad m_2 = \mathbb{E}\left[\tfrac{q}{t}\right], \quad m_3 = \mathbb{E}\left[\tfrac{1}{t^2}\right], \quad m_4 = \mathbb{E}\left[\tfrac{q}{t^2}\right], \quad m_5 = \mathbb{E}\left[\tfrac{q^2}{t^2}\right]. \tag{14}$$

For the premium $X = (P_{od} - q)/t$, the first two moments follow exactly for *any* fare:

$$\mathbb{E}[X] = P_{od}\,m_1 - m_2, \qquad \mathbb{E}[X^2] = P_{od}^2\,m_3 - 2P_{od}\,m_4 + m_5. \tag{15}$$

With b lognormal (median b_0, $s_b^2 = \ln(1 + \mathrm{CV}_b^2)$) and independent of the trip, $\mathbb{E}[1/b] = e^{s_b^2/2}/b_0$ and $\mathbb{E}[1/b^2] = e^{2s_b^2}/b_0^2$, so

$$\mu_{od} = \mathbb{E}[\theta] = \mathbb{E}[1/b]\,\mathbb{E}[X], \tag{16}$$

$$\sigma^2_{od} = \mathbb{E}[1/b^2]\,\mathbb{E}[X^2] - \mu^2_{od}. \tag{17}$$

Fare rounding. Fares are rounded to the nearest \$2.50. A rounding error ε, uniform on $[-1.25, 1.25]$ and independent of t, adds $\mathrm{Var}(\varepsilon)\,m_3$ to $\mathrm{Var}(X)$. The correction is

$$\mathrm{Var}^{\mathrm{corr}}(X) = \mathbb{E}[X^2] - \mathbb{E}[X]^2 - \frac{2.5^2}{12}\,m_3, \qquad \sigma^{2,\mathrm{corr}}_{od} = \sigma^2_{od} - \frac{2.5^2}{12}\,\mathbb{E}[1/b^2]\,m_3. \tag{18}$$

Thin OD pairs. With n_{od} trips in the month, $\mathrm{sd}(X)$ is shrunk toward the period median m_t: $\widehat{\mathrm{sd}}(X) = \big(n_{od}\,\mathrm{sd}(X) + 30\,m_t\big)/(n_{od} + 30)$, with a floor of \$0.01 per minute.

Uniform benchmark (the "average customer"). The benchmark keeps the same travelers but removes route identity. For the exact share, every OD pair gets the trip-weighted mixture of all OD distributions:

$$a^{\mathrm{uni}}(s) = \frac{\sum_{od} D_{od}\,a^{\mathrm{exact}}_{od}(s)}{\sum_{od} D_{od}}. \tag{19}$$

For the DRO share, the pooled moments are $\bar\mu = \sum_{od} D_{od}\mu_{od} / \sum_{od} D_{od}$ and, by the law of total variance, $\bar\sigma^2 = \sum_{od} D_{od}(\sigma^2_{od} + \mu^2_{od}) / \sum_{od} D_{od} - \bar\mu^2$. Both models then have the same average level and differ only in heterogeneity.

Scenarios. Realistic fares give $\theta > 1$ on many OD pairs, so results are reported for the four scenarios of Table 1, not one fare (Figure 5).

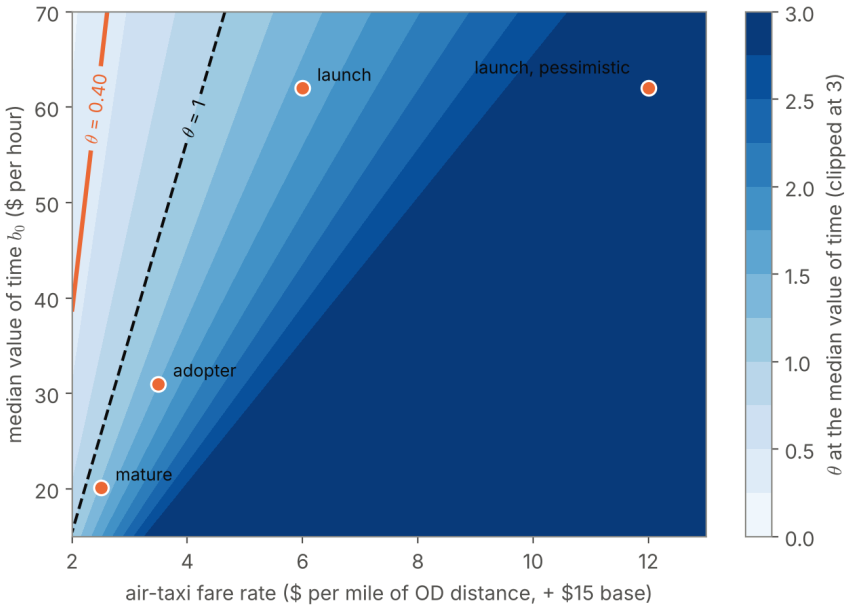


Figure 5: Required saving θ at the median value of time for an illustrative OD pair (35-minute, \$30 rideshare trip, 12 miles), over fare rates and values of time. Dots: the four scenarios of Table 1. Only cheap fares with high values of time bring θ near the paper's 0.40, so Chicago switchers come mostly from travelers above the median value of time.

6 Decision variables

$$y_i = \begin{cases} 1 & \text{if a vertiport is built in zone } i \\ 0 & \text{otherwise} \end{cases} \qquad i \in \mathcal{V}$$

$$x^t_{oijd} = \text{share of the travelers from } o \text{ to } d \text{ in period } t \text{ who are assigned to the vertiport pair } i \to j \qquad (o,d) \in \mathcal{P},\ (i,j) \in \mathcal{K}^t_{od},\ t \in \mathcal{T}$$

Option (o,i,j,d) carries $D^t_{od}\, a^t_{oijd}\, x^t_{oijd}$ air-taxi passengers per day in period t.

7 Optimization models

7.1 Model P0: NYC validation (Table 4 of the paper)

Passengers minus a cost per vertiport:

$$\max \quad \sum_{(o,d)\in\mathcal{P}} \sum_{(i,j)\in\mathcal{K}_{od}} D_{od}\, a_{oijd}\, x_{oijd} \;-\; c \sum_{i\in\mathcal{V}} y_i \tag{20}$$

$$\text{s.t.} \quad \sum_{(i,j)\in\mathcal{K}_{od}} x_{oijd} \le 1 \qquad \forall (o,d) \in \mathcal{P} \tag{21}$$

$$\sum_{j:(i,j)\in\mathcal{K}_{od}} x_{oijd} \le y_i \qquad \forall (o,d) \in \mathcal{P},\ i \in \mathcal{V} \tag{22}$$

$$\sum_{i:(i,j)\in\mathcal{K}_{od}} x_{oijd} \le y_j \qquad \forall (o,d) \in \mathcal{P},\ j \in \mathcal{V} \tag{23}$$

$$y_i \in \{0,1\}, \quad 0 \le x_{oijd} \le 1 \qquad \forall\, i,\ (o,i,j,d) \tag{24}$$

Here c is a cost per vertiport in passenger units, calibrated so that $\mu = 0.40$ yields the paper's three vertiports. μ is then lowered from 0.40 to 0.33 with c fixed (Appendix A).

7.2 Model P1: one period (Chicago)

Profit, or passengers with $\pi = 1$ and $g = 0$:

$$\max \quad \sum_{(o,d)\in\mathcal{P}} \sum_{(i,j)\in\mathcal{K}_{od}} \pi_{oijd}\, D_{od}\, a_{oijd}\, x_{oijd} \;-\; \sum_{i\in\mathcal{V}} g_i\, y_i \tag{25}$$

$$\text{s.t.} \quad \sum_{(i,j)\in\mathcal{K}_{od}} x_{oijd} \le 1 \qquad \forall (o,d)\in\mathcal{P} \tag{26}$$

$$\sum_{j:(i,j)\in\mathcal{K}_{od}} x_{oijd} \le y_i \qquad \forall (o,d)\in\mathcal{P},\ i\in\mathcal{V} \tag{27}$$

$$\sum_{i:(i,j)\in\mathcal{K}_{od}} x_{oijd} \le y_j \qquad \forall (o,d)\in\mathcal{P},\ j\in\mathcal{V} \tag{28}$$

$$\sum_{i\in\mathcal{V}} y_i \le p \tag{29}$$

$$\frac{1}{H} \sum_{(o,d)\in\mathcal{P}} \sum_{(i',j')\in\mathcal{K}_{od}:\ i\in\{i',j'\}} D_{od}\, a_{oi'j'd}\, x_{oi'j'd} \;\le\; Q\, y_i \qquad \forall i\in\mathcal{V} \tag{30}$$

$$y_i \in \{0,1\}, \quad 0 \le x_{oijd} \le 1 \qquad \forall\, i,\ (o,i,j,d) \tag{31}$$

Constraint	Meaning
(26) Assignment	each OD pair uses at most one combination in total; the rest stays on the ground
(27) Departure link	travelers depart only from a built vertiport
(28) Arrival link	travelers arrive only at a built vertiport
(29) Budget	at most p vertiports (all reported runs, $p = 1, \ldots, 10$)
(30) Capacity	departing plus arriving passengers per hour at site i at most Q (H: hours in the period; optional)

7.3 Model P2: one network for all periods (RQ2)

$$\max \quad \sum_{t\in\mathcal{T}} w_t \sum_{(o,d)\in\mathcal{P}} \sum_{(i,j)\in\mathcal{K}^t_{od}} \pi^t_{oijd}\, D^t_{od}\, a^t_{oijd}\, x^t_{oijd} \;-\; G \sum_{i\in\mathcal{V}} g_i\, y_i \tag{32}$$

$$\text{s.t.} \quad \sum_{(i,j)\in\mathcal{K}^t_{od}} x^t_{oijd} \le 1 \qquad \forall (o,d),\ t \tag{33}$$

$$\sum_{j} x^t_{oijd} \le y_i, \qquad \sum_{i} x^t_{oijd} \le y_j \qquad \forall (o,d),\ i \text{ or } j,\ t \tag{34}$$

$$\text{capacity per period (optional)} \qquad \forall i,\ t \tag{35}$$

$$y_i \in \{0,1\}, \quad 0 \le x^t_{oijd} \le 1 \tag{36}$$

G is the number of operating days per year, so both terms are yearly. In the reported runs $\pi = 1$, $g = 0$ and a budget $\sum_i y_i \le p$ is added, so the objective is yearly passengers. The network y is shared by all periods; only the routing x^t changes (Figure 6).

8 Model properties

Proposition 1 (The routing is integral for a fixed network)**.** *For any fixed $y \in \{0,1\}^{\mathcal{V}}$, Model P1 without capacity has an optimal solution in which each OD pair uses at most one vertiport pair,*

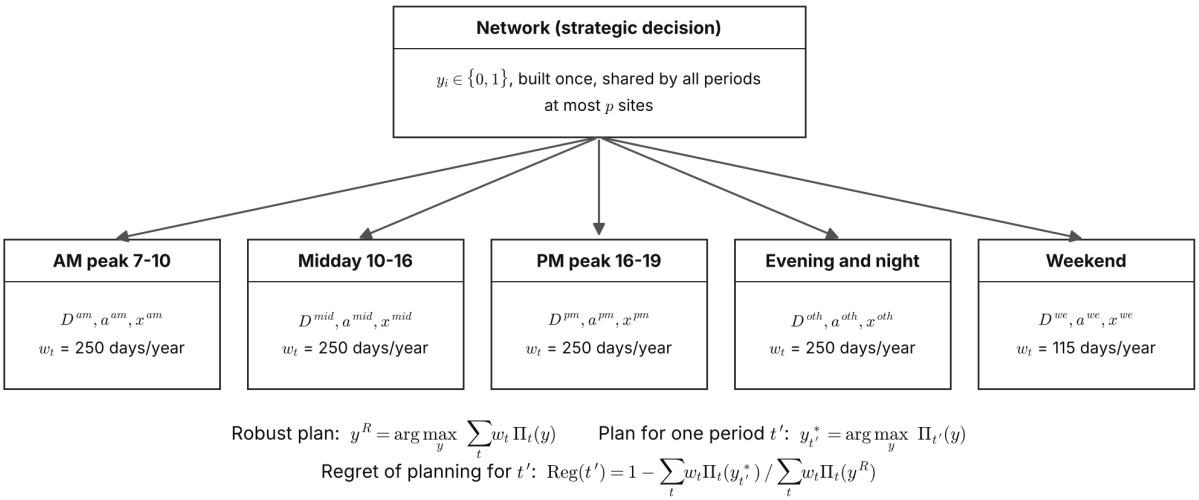


Figure 6: Structure of Model P2 and comparison with single-period plans (passenger objective).

with $x_{oijd} \in \{0,1\}$. *The optimal routing sends OD pair* (o,d) *through*

$$(i^*, j^*) \in \arg\max\{\pi_{oijd}\, a_{oijd} : (i,j) \in \mathcal{K}_{od},\ y_i = y_j = 1\}$$

if that maximum is positive, and keeps it on the ground otherwise.

Sketch. With y fixed, the problem separates by OD pair. The linking constraints force $x_{oijd} = 0$ if $y_i = 0$ or $y_j = 0$; otherwise the assignment constraint implies them. The remaining set $\{x \geq 0 : \sum x \leq 1\}$ is a simplex, so a linear objective peaks at a vertex: all zeros or one option at 1. □

Remark. Proposition 1 gives the value of a network in closed form:

$$\Pi(y) = \sum_{(o,d)} D_{od}\, \max\Big\{0,\ \max_{(i,j)\in\mathcal{K}_{od}:\, y_i = y_j = 1} \pi_{oijd}\, a_{oijd}\Big\} - \sum_i g_i y_i,$$

Any network can thus be evaluated cheaply under any demand scenario, as the metrics of Section 9 require.

Proposition 2 (Strength of the linking constraints)**.**
The aggregated linking constraint $\sum_j x_{oijd} \leq y_i$ *implies* $x_{oijd} \leq y_i$ *for every* j*, but not conversely. The formulation is therefore at least as tight as the classical "strong" facility-location formulation.*

Structure and size. In P1, each customer (OD pair) needs *two* open facilities. This is closer to hub location than to uncapacitated facility location; both are NP-hard in general. The main Chicago instance has 16 binaries and 2,289 options over all periods ($s \geq 0$); 617 weekday options have $s \geq 0.15$. The largest robustness instance has 77 binaries and about 36,000 options. Gurobi solves each instance in seconds. With 16 candidates and $p \leq 8$, all 39,186 networks can also be evaluated exactly with Proposition 1, which reproduces the Gurobi solutions.

9 Evaluation metrics

Let $\Pi^S(y)$ be the value of network y under demand scenario S, with optimal routing (Proposition 1).

Definition 1 (Value of heterogeneity, RQ1)**.** Let y^{het} be optimal with OD-specific shares and y^{unif} optimal with the pooled shares of Section 5. Taking the OD-specific model as the truth,

$$\text{VOH} = \Pi^{\text{het}}(y^{\text{het}}) - \Pi^{\text{het}}(y^{\text{unif}}) \ \geq 0, \qquad \text{VOH\%} = \frac{\text{VOH}}{\Pi^{\text{het}}(y^{\text{het}})}.$$

VOH is the loss from planning with one average expectation. It is reported per scenario, period and budget p.

Definition 2 (Forecast error of the average-customer model, RQ1)**.** For network y^{unif}, the relative gap between the pooled model's prediction and the OD-specific value: $\text{FE} = \big(\Pi^{\text{unif}}(y^{\text{unif}}) - \Pi^{\text{het}}(y^{\text{unif}})\big)/\Pi^{\text{het}}(y^{\text{unif}})$. Negative values are under-forecasts.

Definition 3 (Regret of a single-period plan, RQ2)**.** Let y^R solve P2 and $y^*_{t'}$ solve P1 with period-t' demand only, both with budget p. The regret of planning for t' is the share of yearly passengers lost:

$$\text{Reg}(t') = 1 - \frac{\sum_t w_t\, \Pi_t(y^*_{t'})}{\sum_t w_t\, \Pi_t(y^R)} \;\geq 0.$$

With capacity, $\Pi_t(y^*_{t'})$ comes from re-solving the routing with y fixed.

Network similarity. For networks y and y': the Jaccard index $|S \cap S'|/|S \cup S'|$ of their site sets, and for each site of y' the distance to the nearest site of y. Used to compare with the paper and with Volakakis and Mahmassani (2024).

Benchmark: p-median (not run yet). Choose p sites that minimize demand-weighted access distance,

$$\min \sum_o \sum_i D_o\, \ell_{oi}\, z_{oi} \quad \text{s.t. } \sum_i z_{oi} = 1,\; z_{oi} \leq y_i,\; \sum_i y_i = p,\; y \in \{0,1\},\; z \geq 0,$$

then evaluate the network with $\Pi(y)$. The paper reports 22–53% lower profit for this benchmark.

10 Relation to the original model

Component	**Wang, Jacquillat and Vaze (2022)**	**This project**
Vertiport decisions	location and capacity level z_{ih}, fleet Γ	location y_i (capacity optional)
Matching	y_{oijd} binary, π_{oijd} fractions	x^t_{oijd} (integral at optimum)
Pooling wait	endogenous, $\delta_{ij} \geq (n-1)/\chi_{ij}$ (second-order cone)	fixed $\bar{\delta}$
Detours, reliability	endogenous (queueing network, M/M/1)	not modeled
Fleet, repositioning, charging	modeled	not modeled
Demand function	DRO or MNL S-curve inside the model	exact share (main) and DRO S-curve (comparison), precomputed; optional transfer penalty h
Customer expectations	uniform μ (0.40), σ	**uniform and OD-specific from fares (RQ1)**
Time of day	one demand snapshot (morning commute)	**five periods, robust network (RQ2)**
Solution method	adaptive discretization (MISOCP)	MILP, Gurobi

The simplified model is close to the paper's "exogenous operations" benchmark. The natural next step restores the pooling term: with passenger flow $\chi_{ij} = \sum_{(o,d)} D_{od}\, a_{oijd}\, x_{oijd}$, impose $\delta_{ij} \geq$

$(n-1)/(\chi_{ij}+\varepsilon)$ as a second-order cone constraint (Lemma 1 of the paper). The share a then depends on decisions; a static piecewise-constant discretization (the paper's Sta(M_C) benchmark) keeps the model solvable.

11 Changes in version 1.1

Version 1 predates the Chicago runs. Version 1.1 aligns the document with the code and the reported results:

- the exact share replaces the DRO share in the main results; options need $s \geq s_{\min} = 0.15$ (Section 4.4);
- expectations use a lognormal value of time, the scenarios of Table 1, shrinkage of thin OD pairs and pooled benchmarks (Section 5);
- all runs maximize passengers under a budget of p vertiports; regret is a share of yearly passengers; a forecast-error metric is added (Section 9);
- the transfer penalty, capacity, bootstrap and seasonal check below are new.

11.1 Transfer penalty

Like the paper's equations (19)–(20), the utility of Section 4.4 has no fixed cost for changing vehicles. The paper's survey-based expectations absorb such costs implicitly; data-based expectations do not. The penalty h enters τ and takes values $\{0, 5, 10, 15\}$ minutes, with $s_{\min} \in \{0.15, 0\}$. Flight overhead and pooling wait add fixed minutes to τ in the same way, and cruise speed nearly so, so this analysis also bounds their effect.

11.2 Vertiport capacity

Constraint (30) is applied in P2 for every period, with H_t = 3, 6, 3, 12 and 24 hours (AM, midday, PM, other weekday hours, weekend). Demand is spread evenly within each period, and $Q \in \{\infty, 200, 100, 50\}$ passengers per hour (departing plus arriving). Capacity breaks the closed form of Proposition 1, so single-period networks are evaluated by re-solving the routing with y fixed. The values of Q are illustrative.

11.3 Exact evaluation by enumeration and day-level bootstrap

Without capacity, Proposition 1 values every network in closed form. With 16 candidates and $p \leq 8$ there are 39,186 networks, all evaluated directly; this reproduces the Gurobi solutions of P1 and P2 exactly. The 22 weekdays and 9 weekend days of March 2026 are resampled with replacement 200 times, separately by day type. Each resample rebuilds demand, ground times, options and shares, then re-evaluates every network. Reported intervals are the 5th and 95th percentiles. They capture sampling variation within March-like conditions, not uncertainty in assumed parameters.

11.4 Seasonal check

The same pipeline runs on July 2025 and January 2026. For each p, the network planned for one month is evaluated on the other two. An all-season network maximizes the sum of the three monthly values, each divided by its own optimum (equal month weights).

12 Runs reported

Run	Content	Model
V1	NYC validation: μ swept from 0.40 to 0.33 with a calibrated cost c per vertiport (Appendix A)	P0
C1	RQ1: OD-specific vs pooled shares, four fare scenarios, five periods and weekday, $p = 1, \ldots, 10$; 16 sites and all 77 areas ($k = 3$ and $k = 8$)	P1
C2	RQ2: robust network vs single-period networks, three scenarios, $p = 2, \ldots, 8$; 16 sites and 77 areas	P2, P1
C3	Capacity $Q \in \{\infty, 200, 100, 50\}$, adopter scenario	P2
C4	Day-level bootstrap, 200 resamples, three scenarios	enumeration
C5	Transfer penalty $h \in \{0, 5, 10, 15\}$, $s_{\min} \in \{0.15, 0\}$	enumeration
C6	Seasonal check: July 2025, January 2026, March 2026	enumeration

Not yet run: p-median benchmark; spread of the value of time (CV_b from 0.5 to 1.2); March 2019 comparison; endogenous pooling.

A NYC validation results (Model P0)

Yellow taxi, March 2019, weekdays 7–10 am, demand scaled by $S = 2$; 26 candidate zones; $v = 240$ km/h, $t_{\mathrm{oh}} = 10$, $\bar{\delta} = 5$, $\sigma = 0.20$; $c = 36.75$ passengers (any $c \in [32, 41.5]$ gives 3 vertiports at $\mu = 0.40$).

μ	Sites (ours)	Sites (paper)	Passengers	Change in our network
0.40	3	3	125.6	JFK, Times Sq/Theatre District, Gramercy
0.39	3	3	136.6	–
0.38	4	3	187.0	+ Upper East Side North
0.37	4	4	201.9	–
0.36	6	4	290.3	+ Kips Bay, LaGuardia, TriBeCa; − Gramercy
0.35	6	7	316.9	–
0.34	7	9	382.8	+ Gramercy, Midtown Center; − Kips Bay
0.33	7	10	416.7	–

The number of vertiports tracks the paper. Most site differences are adjacent zones (Times Sq vs Clinton East). The model puts the LaGuardia-area vertiport at the airport; the paper puts it in Astoria for operational reasons: shorter flights, higher reliability, shorter detours (Section 7.1.1 of the paper). This gap isolates the effect of the operations layer omitted here.